End Semester Examination -2023

Course: B.Tech. | *Semester-IV* | *Time: 03 Hours*

Code: 101406 | *Subject: Materials, Testing and Evaluation* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct answer of the following (any seven Question only):* [2 x 7 = 14]

(a) The rapid hardening Portland cement has a
- (i) Lower heat of hydration
- (ii) Higher heat of hydration
- (iii) Lower shrinkage coefficient
- (iv) Higher shrinkage coefficient

(b) The purpose of seasoning of timber is to
- (i) Change the direction of grains
- (ii) Remove voids
- (iii) Reduce moisture content
- (iv) All of these

(c) The chief ingredient of a paint is
- (i) Alcohol
- (ii) Drier
- (iii) Oil
- (iv) Pigment

(d) Which of the following is not a zone of fatigue fracture?
- (i) Zone of crack nucleation
- (ii) Fatigue Zone
- (iii) Elastic zone
- (iv) Final fracture

(e) The slope of the stress-strain curve in the plastic range is
- (i) Bauschinger effect
- (ii) Resolved shear stress
- (iii) True strain
- (iv) Rate of strain hardening

(f) Which of the following is an example of a thermo plastic?
- (i) Melamine
- (ii) Epoxide
- (iii) Urethane
- (iv) Acetal

(g) Which of the following is a characteristic of asbestos minerals?
- (i) Heat resistance
- (ii) Reacts with acids
- (iii) Poor strength
- (iv) Poor bonding

(h) Which of the following is not a method of fabrications of rubber?
- (i) Vacuum forming
- (ii) Buffing
- (iii) Calendaring
- (iv) Compression molding

(i) Fire brick is an important row material of refractory metals, which is made from
- (i) Brick
- (ii) Concrete
- (iii) Fire clay
- (iv) Wood

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(iii) 96% silica (iv) Borosilicate

Q.2 (a) Enumerate the laboratory tests for cement and describe any one of them. [7]
(b) What are the qualities of good timber? Explain in detail. [7]

Q.3 (a) What are the characteristics of an ideal paint? Discuss briefly. [7]
(b) Differentiate between cup shakes and heart shakes with neat sketches. [7]

Q.4 (a) What are the various market forms of rolled sections steel. Discuss each with neat sketches. [14]

Q.5 (a) Table 1 shows the details of weights of sample retained on various sieves. The total weight of the coarse aggregate is 10 kg and the fine aggregate is 1 kg. Determine the fineness modulus of the coarse and dine aggregate. [7]

Table 1 : details of weights of sample

Sieve	Coarse aggregate weight retained (in kg)	Fine aggregate weight retained (in gm)
80 mm	0.2	-
40 mm	0.4	-
20 mm	3.2	-
10 mm	3	0
4.75 mm	3.2	25
2.36 mm	-	130
1.18 mm	-	140
600 µm		350
300 µm		250
150 µm	-	105

(b) Define maturity of concrete. How the compressive strength of concrete varies with maturity? Explain segregation. Bleeding and laitance in green concrete. Explain elongation index and flakiness index of aggregates. [7]

Q.6 Attempt any one of the following: [14]
(a) Discuss the various and uses of plastics in the construction industry.
(b) Explain the properties and uses of glass as a construction material.

Q.7 (a) Explain the factors governing the choice of Building Material. [7]
(b) Explain the various constituents of paint. [7]

Q.8 (a) Discuss about factors affecting the Acoustic design of buildings. [7]
(b) What are the various forms of bitumen. Discuss in detail. [7]

Q.9 Write short notes on ***any four*** of the following: [3½ x4 = 14]
(a) Engineering stress strain curve and true stress strain curve.
(b) Refractories and graphene
(c) Creep and fatigue
(d) Structural integrity
(e) Torsion test of steel
(f) Properties of asbestos.

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